TISSUE

- **Tissue** is a group of cells that are similar in structure and/or work together to achieve a particular function.
- Example of tissues: Blood, phloem and muscle
- In tissues the cells are arranged and designed so as to give the highest possible efficiency of function.
- **Utility of tissues in multicellular Organisms:** Tissues give structural and mechanical strength and facilitate the division of function in multicellular organisms. For example, nerve cells from the nervous tissue aid in message transmission and muscle cell from muscle tissue act as per the message of nervous tissue.
- **Comparison between Plant Tissue and Animal Tissue:**

	Plant Tissue	Animal Tissue
1	Plants have large quantity of supportive tissues since they have to be upright and fixed.	Animals have less supportive tissues as animals have to move.
2	They have more dead cells (dead cells are mainly supportive cells).	Animals generally have living cells as they have to function with whole body.
3	Plants have tissues for growth only in certain parts of it which divides throughout their life as a few parts of plants must always grow.	Animals have tissues for the growth of overall body and not just certain parts as animals need growth in overall body.
4	Cells of plant tissues stores more of sugars as need more of cellulose.	Cells of animal tissues synthesize and stores more of protein because animal body requires more of protein.
5	Requires less energy for maintenance.	Requires more energy for maintenance.
6	Cells of plant tissues have cell walls.	Cells of animal tissues do not have cell walls.
7	Most of the tissues are autotrophs.	Most of the tissues are heterotrophs.
8	Tissues of plants cells generally have chloroplast.	Tissues of animal cells do not have chloroplast.

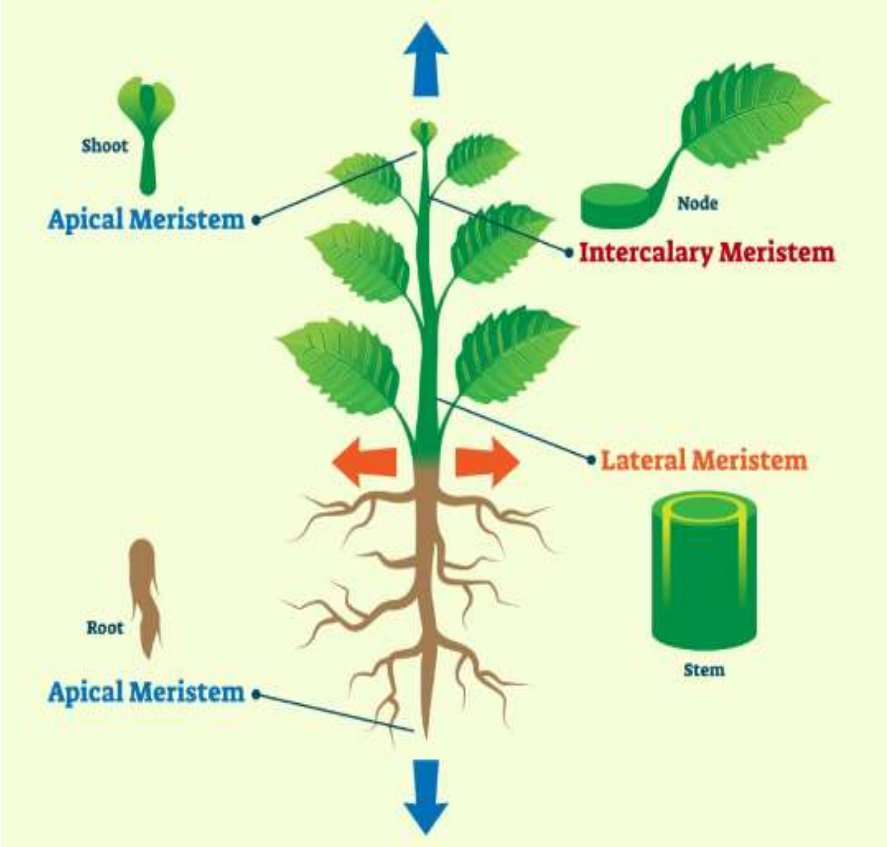
Plant Tissues

- ✓ Plant Tissues can be divided into two categories: **Meristematic Tissue** and **Permanent Tissue.**
- ✓ Dividing capacity is the basis of classifying various plant tissues.
- ✓ **Meristematic Tissue:**
  - These tissues are responsible for plant growth.
  - Meristematic Tissues are present at some specific locations where the growth of plant occurs.
  - Meristematic Tissues are made up of undifferentiated cells (which do not have a specialized function).
  - **Characteristics of cells of Meristematic Tissue**:
    - Cells of meristematic tissue are very active(living and dividing).
    - They have dense cytoplasm, thin cellulose walls and prominent nuclei to help in cell division.
    - They lack vacuoles because vacuoles provide rigidity and turgidity to cell which can cause problem in cell division.
  - New cells produced by meristem are initially like those of meristem itself, but as they grow and mature, their characteristics slowly change and they may become components of other tissues.
  - Meristematic tissues are classified as Apical, Lateral and Intercalary on the basis of the region where they occur.
  - **Apical Tissue:**
    - **Location:** Growing tips of stem and roots.
    - **Function:** Increases the length of stem and roots.
    - On the basis of location, Apical Meristem is categorized into two parts: **Shoot Apical Meristem(SAM)** and **Root Apical Meristem(RAM).**

Conceptual Learning

- **Lateral Meristem:**
  - **Location:**lateral side of stems and roots
  - **Function:**Increases the thickness or girth of stem and roots
  - **Example:**VascularCambium of plant is also a Lateral Meristem.
  - Example of plants in which lateral meristem is not significant is Palm tree.

- **Intercalary Meristem:**
  - **Location:** Near the nodes of plants especially the base of leaves.
  - **Function:** Help in increasing the length of the internodes.
  - Intercalary is not significant in dicots plants like pea and legumes.
  - It is present in grass, plant stem at the base of nodes and leaf blades.



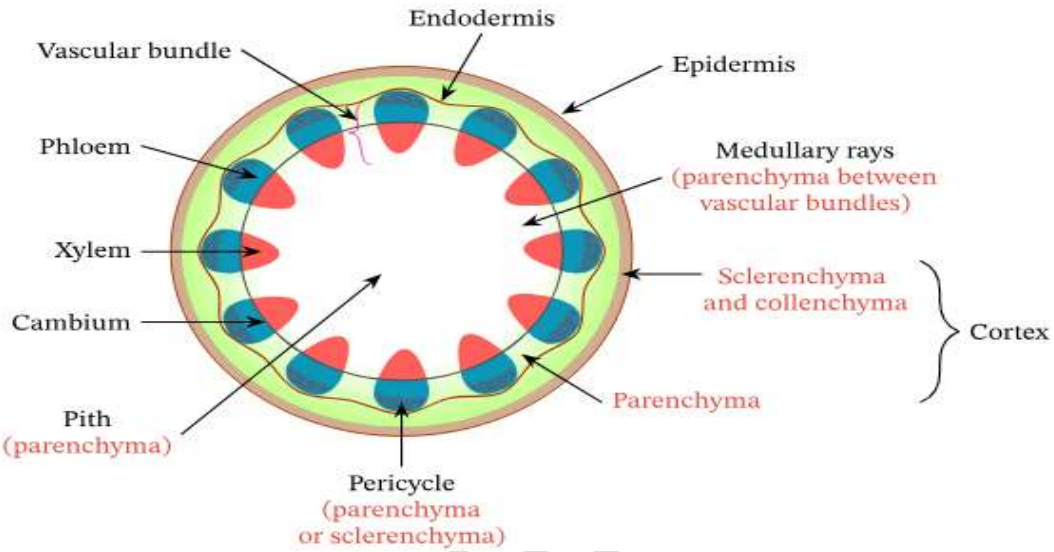
✓ **Permanent Tissue:**

- Permanent tissues are a group of living or dead cells which are formed by meristematic tissues that have lost their ability to divide.
- Permanent tissues are made up of cells that do not undergo cell division.
- Meristematic tissue becomes various types of permanent tissues at a certain stage by the process of Differentiation.
- **Differentiation** is the process in which the cells of meristematic tissue take up a specific function and lose the ability to divide.
- **Functions of Permanent tissue:**
  - It provides support to plant.
  - It gives protection to plant.
  - It helps in photosynthesis.
  - It helps in conduction of water, minerals and nutrients.
- **Characteristics of cells of Permanent Tissue:**
  - Cells of permanent tissue do not undergo cell division.
  - Cells of the permanent tissue are derived from the meristematic tissue.
  - Cells of permanent tissues are fully differentiated.
  - Cells of permanent tissue have comparatively large shape and size.
  - Cells of permanent tissues have large vacuoles.
  - Cells of permanent tissues have high metabolic rate.
  - Cells of permanent tissues may be dead or living.
- On the basis of structure and function of cells, Permanent Tissues can be classified into two parts: **Simple Permanent tissues** and **Complex permanent tissues.**

Conceptual Learning

Simple Permanent tissues:

- **Simple permanent tissues** are the type of permanent tissues that are made of similar types of cells that perform the same function.
- This tissue is majorly found in the construction of organs, beneath the epidermal layer and the parts where mechanical support is provided.
- Generally, simple permanent tissues are made up of only one type of cells.
- On the basis of function, Simple Permanent tissues are classified into three types: **Parenchyma, Collenchyma** and **Sclerenchyma.**

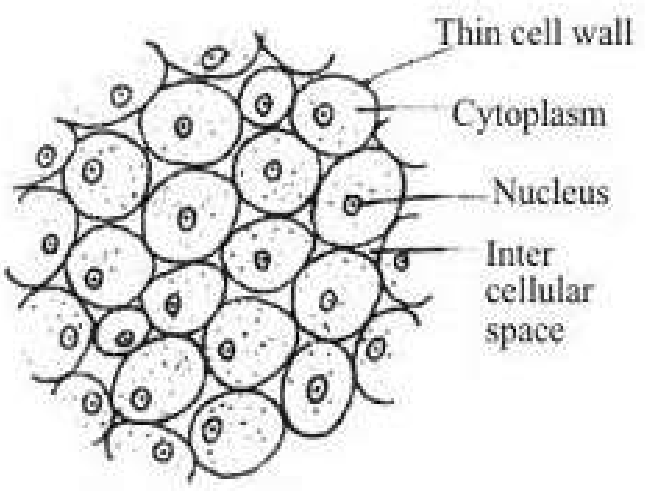


Cross Section of Stem

- **Parenchyma:**

⇒ **Parenchyma** is a type of simple permanent tissue that makes a major part of ground or fundamental tissues in plants, where other tissues like vascular tissues are embedded.

⇒ **Location:** The parenchyma tissues are found throughout the plant. It's also found in the plant's soft portions, such as the cortex of roots, ground tissues in stems, and leaf mesophyll.



⇒ Parenchyma is the most common simple permanent tissue.

⇒ **Characteristics of Cells of Parenchyma:**

- It consists of relatively unspecialized cells.
- The cells have thin walls.
- The cells are living.
- The cells are loosely arranged.
- Cells have large intercellular spaces between cells because of loose arrangement.
- Cells contains large central vacuoles.

⇒ **Functions of Parenchyma:**

- This tissue generally stores food.
- They are the hub of all metabolic activity.
- It helps in photosynthesis and respiration.

- Supports and serves as a foundation of plant life. A single parenchyma cell of a zygote can develop into an entire plant.
- Helps in in wound regeneration, healing, and repair.

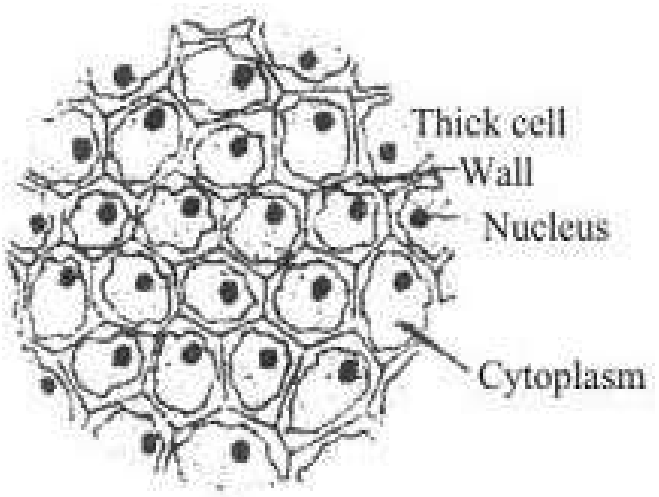
⇒ **Special Parenchyma:**

1) **Chlorenchyma:** It is the type of parenchyma tissue in which cells contains chlorophyll and performs photosynthesis. The stem and the aerial root of the plants are green due to the presence of Chlorenchyma.
2) **Aerenchyma:** It is the type of Parenchyma tissues which contains large air cavities. Parenchyma is generally found in aquatic plants and it helps them to float. The Aerenchyma naturally forms in the root of rice. These cells store respiratory gases (oxygen and carbon dioxide), which can be utilized by the plants when required.

- **Collenchyma:**

⇒ **Collenchyma** is a type of plant tissue made up of living cells that are elongated and have thickened cell walls**.**

⇒ **Location:** It is generally found in flexible parts of plants like tendrils, leaf stalks below epidermis, stems of climbers etc.



⇒ **Characteristics of Cells of Collenchyma:**

- The cells of this tissue are living.
- The cells are elongated
- The cells are irregularly thickened at the corners.
- There is very little intercellular space

⇒ **Functions of Collenchyma:**

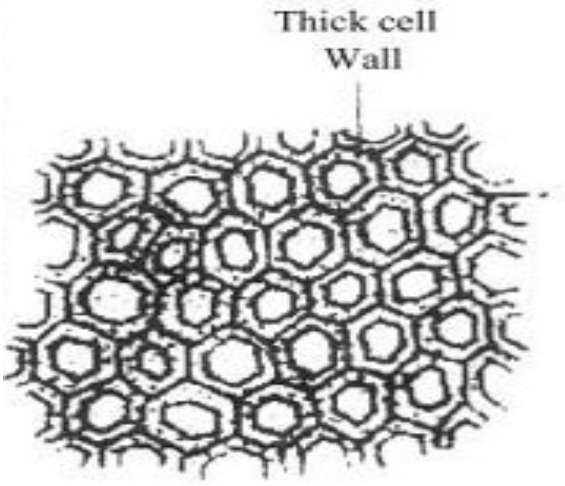
- It provides flexibility to the plant.
- It allows various parts of plant to bend.

- **Sclerenchyma:**

⇒ **Sclerenchyma** is a type of plant tissue made up of dead cells and provides hardness in various parts of plants.

⇒ **Location:** It is generally found in stems, around vascular bundles, in the veins of leaves and in the hard covering of seeds and nuts.

⇒ It is impervious to water because of its thick and strong lignin-based outer cell walls.



⇒ **Characteristics of Cells of Sclerenchyma:**

- The cells are dead.
- The cells are long and narrow as the walls are thickened due to lignin (kind of cellulose).
- The cells have thick cell walls and, in some cases, intercellular space becomes negligible because of it.

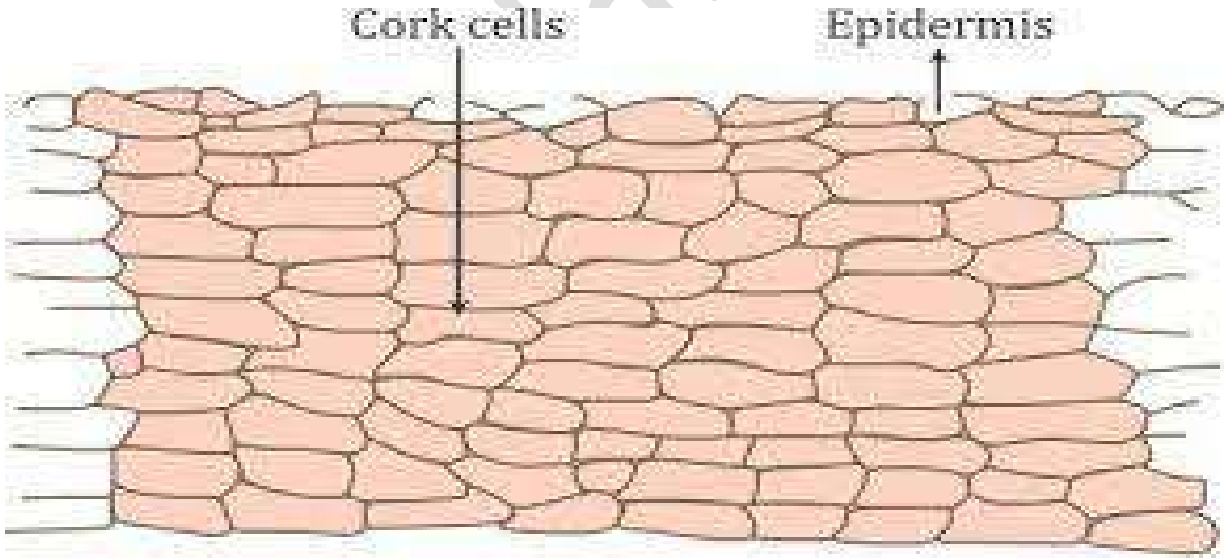
⇒ **Functions of Sclerenchyma:**

- It provides strength to the plant parts.
- It provides mechanical support to water.

Conceptual Learning

Plant Protective Tissue:

- The are two kinds of plant protective tissues:
  - Plant Epidermis is the outermost protective tissue that covers the entire surface of the plant.
  - The epidermis is usually made of a single layer of cells.
  - The epidermis of plants in dry areas may be thicker to minimize water loss.
  - Cells of Epidermis are called as **Epidermal cells.**
  - Stomata are the small pores in the epidermis of leaf. Guard cells are the cells in Epidermis which encloses and regulates the opening and closing of stomata.
- **Location of Epidermis:** Entire surface of plant.
- **Characteristics of cells of Epidermis:**
  - Cells form a continuous layer without intercellular spaces to protect the plant.
  - Cells are living.
  - Most of epidermal cells are relatively flat.
  - Outer and sidewalls of the epidermal cell is thicker than the inner wall.
  - Epidermal cells of roots have long hair-like parts to increase area of water absorption.
- **Functions or Epidermis:**
  - It protects all the parts of Plant from mechanical injury.
  - It protects plants from water loss. Epidermal cells on aerial parts of plant secrete a waxy and water-resistant layer on outer surface to minimize water loss. The secretion is called **Cutin.**
  - It protects the plants from parasitic attack.
  - Stomata of Epidermis helps in exchange of gases and transpiration of water.
  - Epidermal cells in roots absorbs water from soil.



Protective tissue

- **Cork:**
  - Cork is the outermost layer of tissue in woody plants that is resistant to the passage of water vapor and gases and that becomes the bark.
  - As the plant grows old, a strip of secondary(lateral) meristem located in the cortex forms layers of cells which constitute the cork.
  - Cells of cork are dead and compactly arranged without intercellular spaces.
  - They also have a substance called **suberin** in their walls that makes them impervious to gases and water.

Conceptual Learning

Complex permanent tissue:

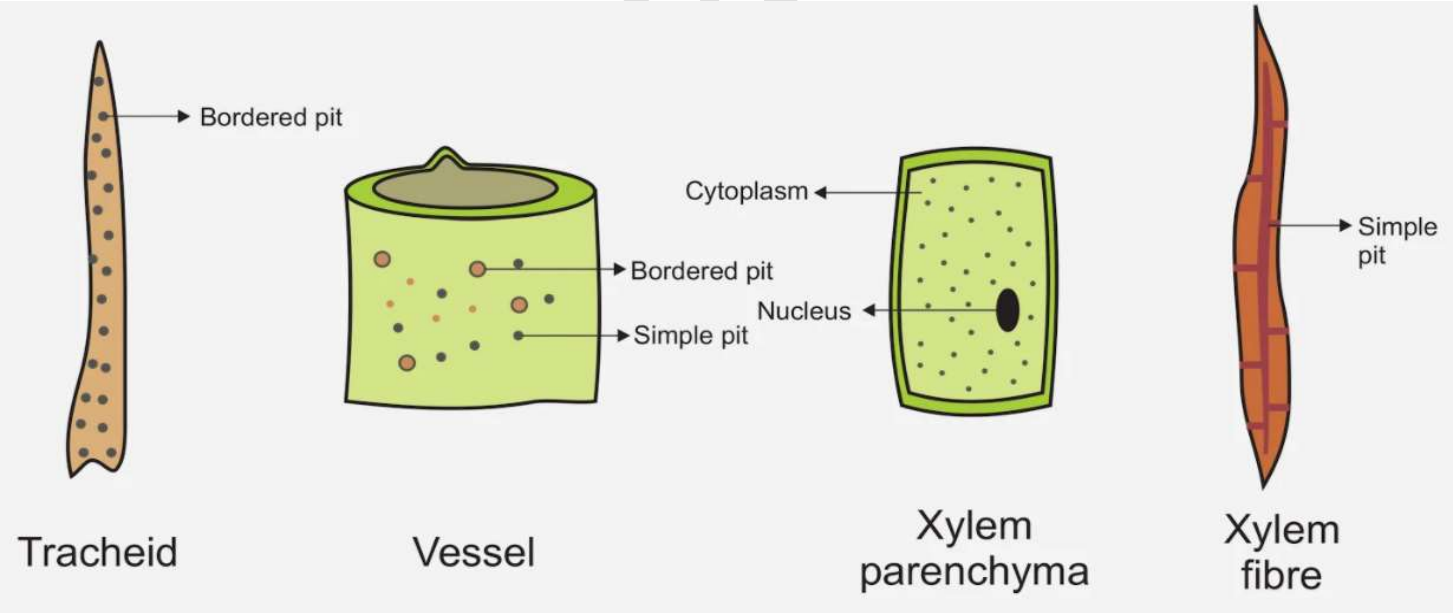
- Complex permanent tissues are the tissues in plants which are made of more than one type of cells to perform a common function.
- These tissues are not growth oriented tissues.
- **Xylem** and **Phloem** are two types of complex permanent tissue.
- Xylem and Phloem are both conducting tissues and constitute a vascular bundle.
- These tissues are also termed as Vascular tissues.
- Vascular tissue is a distinctive feature of the complex plants, one that has made possible their survival in the terrestrial environment.
- Apart from conduction, these tissues also provide mechanical strength to the plant.
- **Xylem:**

⇒ **Xylem** is the conducting tissue responsible for water transport throughout a plant body from the soil via roots up to the stem and leaves.

⇒ **Location of Xylem:** Xylem tissue is found throughout the body of plant.

⇒ **Components/Elements of Xylem:**

- Xylem is composed of mainly four types of elements in which some of them are living and some of them are dead.
- These types are: Tracheids, Vessels, Xylem parenchyma and Xylem fibres.
- Amongst these 4 types, only vessels and tracheids are utilised for sap transport and are together known as tracheary elements.
- Tracheids and vessels are tubular structures to transport water and minerals vertically.



⇒ **Tracheids:**

- These are the fundamental cells of the xylem.
- They have thick walls.
- Trachieds are dead when mature.
- They are elongated with a tube-like structure having tapering (gradual reduction in size) in the end.
- A major portion of the tracheid's cell wall is imperforated.

⇒ **Vessels:**

- Vessels are advanced conducting elements which work along with tracheids for the transportation of xylem sap.
- Vessels consist of a long, cylindrical tube- like structure made up of many cells called vessel members.
- They have thick walls.
- Vessels cells are generally dead.
- Cells of vessels have a central cavity and are surrounded by thickened walls formed due to lignin deposition.
- Vessels have perforated which are present on the end walls of vessel cells.

⇒ **Xylem parenchyma**

- Xylem parenchyma is living cells with thin walls of cellulose.
- Its function is to store foods like fat and starch.
- It helps in lateral movement of water.

⇒ **Xylem Fibres**

- These are elongated dead cells with sharp ends and thick cell walls made up of lignin.
- It provides mechanical strength to the Xylem Tissue.

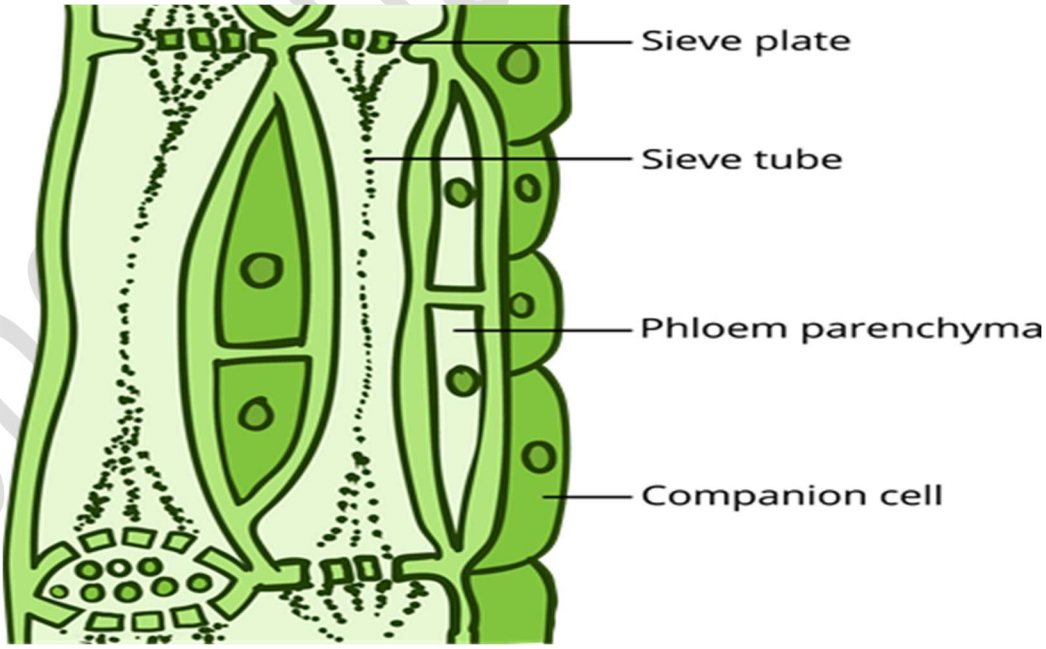
- **Phloem:**

⇒ Phloem is the complex permanent tissue which transport food in plant from the part where it is prepared to other parts.

⇒ The process of transport of food from the parts where it is prepared to other parts is known as **phloem translocation.**

⇒ Phloem translocation is an active process in which plant utilizes its energy.

⇒ Phloem has five elements: **sieve cells, sieve tubes, companion cells, phloem fibres** and **the phloem parenchyma.**



⇒ **Sieve Tubes:**

- Sieve tubes are an advanced type of conducting structures in phloem.
- Sieve tubes are long, tubular cells with perforated (small holes in it) walls.
- Phloem transports food from leaves to other parts of the plant.
- Their cell wall is made up of cellulose.

Conceptual Learning

- Except phloem fibres, other phloem cells are living cells.
- Cells are living.
- Sieve tubes are considered living cells without a nucleus and ribosome because the nucleus of companion cells controls their functional activities.
- Their function is to conduct food.
- Sieve tubes have sieve plates which help in transport of food.

⇒ **Sieve Cells**

- These are the primitive types of conducting elements in the phloem
- These are long, conducting cells in the phloem that do not form sieve tubes.
- Sieve cells do not have sieve plates which help in transport of food.

⇒ **Companion Cells**

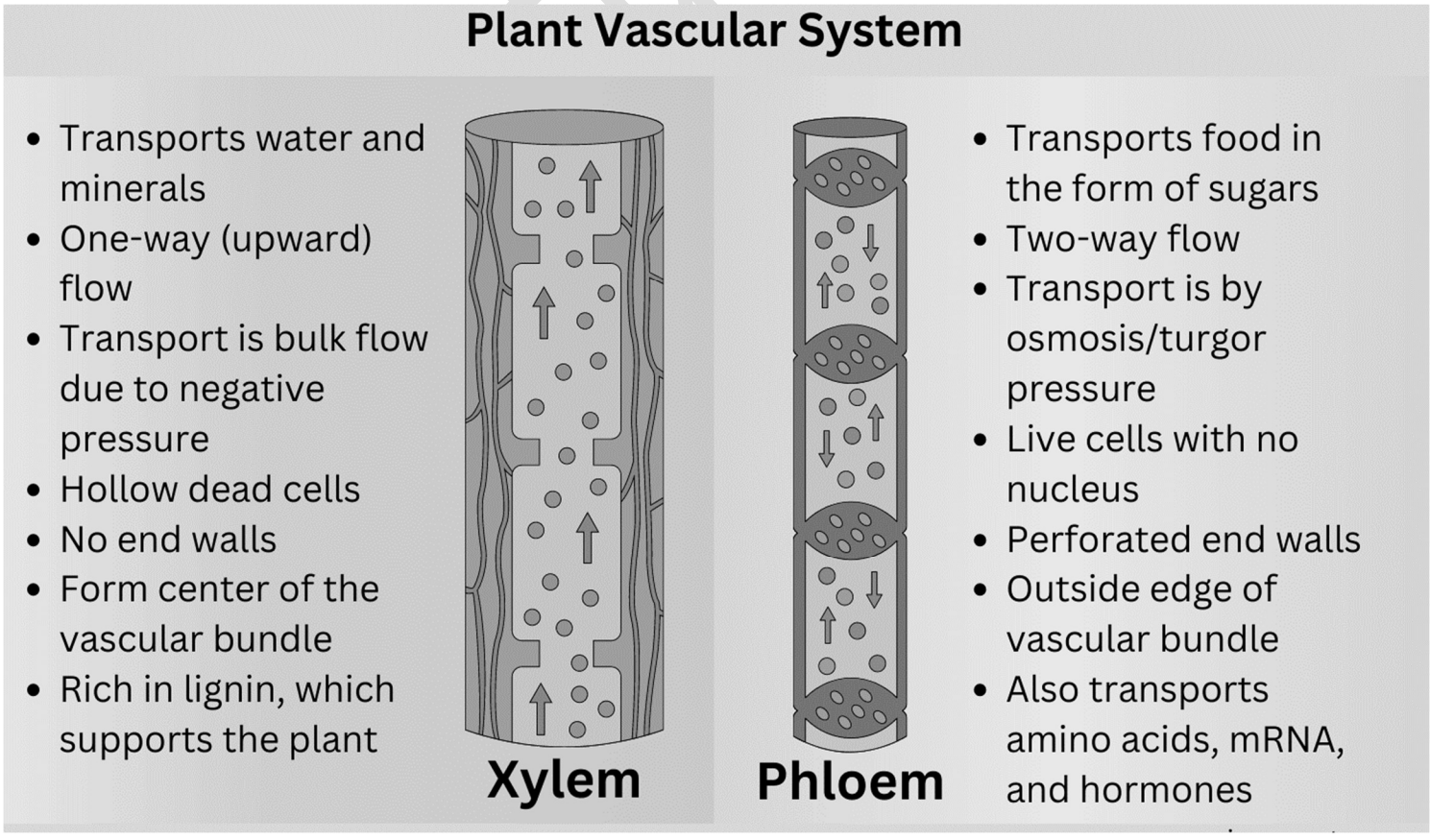
- These are specialized type of parenchyma cells present in phloem tissue.
- These are living cells.
- These are small elongated cells with thin imperforated cell walls.
- Companion cell and sieve tubes function together for the translocation of food.
- The sieve tubes and companion cells are sister cells because it is originated from the same mother cell.

⇒ **Phloem Parenchyma**

- Phloem parenchyma is living cells with thin walls of cellulose.
- Its function is to store food and it helps in sideways conduction of food.
- It is generally present in dicots.

⇒ **Phloem Fibres**

- These are dead cells with thick walls made up of lignin.
- It provides mechanical strength to the Phloem Tissue.
- It is also called **Bast fibre** or **skin fibre.**



Final Learning

Animal Tissues

- ✓ Animal Tissues are classified on the basis of their function.
- ✓ Animal tissues can be classified into four types: **Epithelial Tissue, Muscular tissue, Connective tissue** and **Nervous tissue.**
- ✓ **Epithelial Tissue:**
  - Epithelial Tissue a type of body tissue that forms the covering on all internal and external surfaces of your body, lines body cavities and hollow organs
  - Epithelial tissue covers out outer body as well as inner organs and cavities.
  - It is the major tissue in Glands.
  - **Characteristics of cells of Epithelial Tissue:**
    - Epithelial tissue cells are tightly packed and form a continuous sheet.
    - They have only a small amount of cementing material between them.
    - They have almost no intercellular spaces.
    - All epithelium is usually separated from the underlying tissue by an extracellular fibrous basement membrane.
  - **Functions of Epithelial Tissue:**
    - It protects outer body as well as inner organs.
    - Epithelial tissue works as a barrier to keep organ systems separate from each other.
    - As epithelial tissue is out outer skin, thus anything entering or leaving the body must cross at least one layer of epithelium.
    - Epithelia play an important role in regulating the exchange of materials between the body and the external environment and also between different parts of the body. Material exchange takes place by the permeability of the cells of epithelial tissue.
    - On the basis of structure and function, Epithelial tissues are categorized into various types:

      ⇒ **Simple Squamous epithelial Tissue:**
      - Simple squamous epithelium is a simple flat kind of epithelial tissue.
      - Simple squamous epithelial cells are extremely thin and flat and form a delicate lining.
      - Cells are closely packed.
      - Wherever transportation of materials occurs through a selectively permeable surface, simple squamous epithelium is provided.
      - Examples: (i) linings of blood vessels (ii) Lung alveoli (iii) Linings of mouth cavities

      ⇒ **Stratified Squamous epithelial Tissue:**
      - Stratified squamous epithelium is a epithelial tissue in which there are many layers of squamous.
      - There are no intercellular spaces.
      - It gives protection against water loss.
      - It gives protection against microorganisms.
      - They helps to prevent wear and tear.
      - Example: (i) Tissues of skin (ii) Linings of esophagus (iii) Inner linings of Vagina

      ⇒ **Simple Cuboidal Epithelial Tissue:**
      - Cuboidal epithelial tissue is a type of animal tissue that is made up of a single layer of cube-shaped cells.
      - Cells have cells which have large, spherical and central nuclei.
      - Cells of Cuboidal Epithelial tissue performs the function of secretion and filtration and mechanical support.
      - Example: (i) lining of kidney tubules (ii) ducts of salivary glands (iii) Linings of ovaries

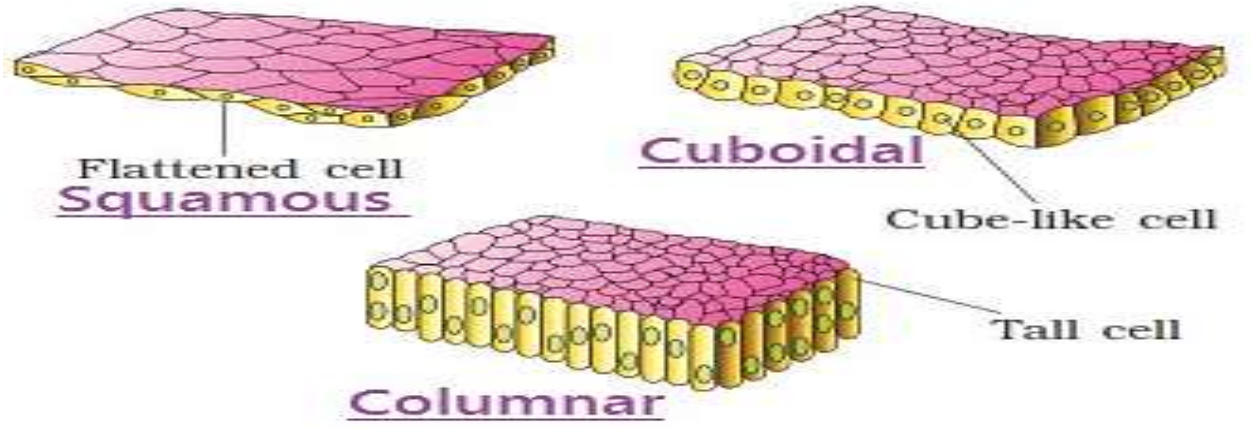
Conceptual Learning

⇒ **Stratified Cuboidal Epithelial Tissue:**

- Stratified cuboidal epithelium is a rare type of epithelial tissue that's made up of multiple layers of cube-shaped cells.
- The most superficial layer is made up of cuboidal cells, while the other layers can be made up of other types of cells.
- The cells in the most superficial layer have round nuclei located in the basal part of the cell.
- Its main function is structural reinforcement
- They helps in protection of ducts.
- Examples: (i) Ducts of sweat glands (ii) Ducts of mammary glands (iii) Ducts salivary glands (iv) Linings of urethra.

⇒ **Simple Columnar Epithelial Tissue:**

- Simple columnar epithelium is a single layer of tall, thin cells that line many organs in the human body.
- Meaning of columnar is pillar-like.
- The cells have oval-shaped nuclei located at the base of the cell.
- Simple columnar epithelium lines most organs of the digestive tract.
- They may be ciliated and non-ciliated depending on their function.
- They perform the function of secretion, absorption and protection.
- Example: (i) Linings of Stomach (ii) Linings of Intestine (iii) Linings of Uterus.



⇒ **Stratified Columnar Epithelial Tissue:**

- Stratified columnar epithelium is a rare type of epithelial tissue composed of column-shaped cells arranged in multiple layers.
- The nuclei of the stratified columnar epithelium are located at the bottom of the cells.
- It's often found between squamous epithelia and simple columnar epithelia.
- The cells function in secretion and protection.
- Examples: (i) Linings of Anus (ii) Male urethra and Vas-deferens (iii) Fetal esophagus (iv) Parts of Pharynx (v) Also occurs in embryo.

⇒ **Pseudostratified columnar epithelium:**

- Pseudostratified columnar epithelium is a tissue that appears layered but is actually made up of a single layer of columnar epithelial cells that are different heights and have their nuclei located in different positions.
- The cells are tightly packed together.
- Pseudostratified epithelia function in secretion or absorption.
- Examples: (i) Internal parts of Ear (ii) Trachea (iii) Bronchi

⇒ **Ciliated Epithelial Tissue:**

- It is a type of epithelial tissue that has hair-like projections called cilia on its outer surface.
- Generally columnar epithelial tissues are found to have cilia.
- The nucleus of the cell is generally oval in shape and found at the base.
- The free ends of the cells sometimes have a brush-like border or cilia.
- It has function of secretion, absorption and movement of other substances.
- Examples: (i) Respiratory Tracts (secretion and movement of mucus) (ii) Fallopian tube (cell movement towards uterus.

⇒ **Glandular Epithelial Tissue:**

- A type of tissue that lines certain internal organs, produces and releases substances in the body, such as mucous, digestive juices, and other fluids.
- Cells of Glandular tissue are modified epithelial cells.
- There main function is secretion.
- Examples: (i) Breast linings (ii) Sweat Glands linings (iii) Tear glands linings

⇒ **Transitional Epithelial Tissue:**

- Transitional Epithelial Tissue a specialized type of stratified epithelium that's made up of multiple layers of cells that can flatten when stretched.
- Transitional epithelium is also known as urothelium.
- It can appear to be stratified cuboidal when the tissue is not stretched.
- it can appear to be stratified squamous when the tissue is stretched.
- Example: (i) urinary bladder lining (ii)nasal cavity lining

✓ **Connective Tissue:**

- Connective Tissue is an animal tissue which supports, protects and links other tissues and organs in the body.
- Connectives tissues are the most abundant tissues found in the body.
- **Components of connective tissue:** Connective tissues have three components:
  i. Intercellular matrix(Ground Substance)
  ii. Cells of connective tissue
- **Characteristics of Intercellular Matrix:**
  - The matrix may be jelly like, fluid, dense or rigid.
  - The nature of matrix differs in concordance with the function of the particular connective tissue
- **Characteristics of cells of Connective Tissue:**
  - The cells of connective tissue are loosely spaced.
  - Embedded in an intercellular matrix.
  - Cells may be stationary or migrating.
- **Functions of Connective Tissue:**
  - Connects various body parts.
  - Prevents organs from getting displaced by body movements.
  - It acts as binding, supporting and packing material.
- **Types of Connective Tissue:** There are following types of connective tissue:
  i. Areolar Connective Tissue
  ii. Adipose Tissue
  iii. Specialized Connective Tissue
  iv. Blood

Conceptual Learning

Areolar Connective Tissue:

- Areolar Connective Tissue is a type of connective tissue having the fibers loosely arranged in a meshwork.
- They are also called as loose connective tissue.
- It is the most abundant tissue.
- It spreads throughout the human body.
- Areolar tissue is tough, yet flexible, and comprises membranes.
- **Types of Areolar Tissue:** Areolar Tissue has two types:
  i. Loose Areolar connective Tissue
  ii. Dense Areolar Connective Tissue
- **Loose Areolar Connective Tissue**

  ⇒ Areolar Tissues are generally called as Loose Connective Tissues.

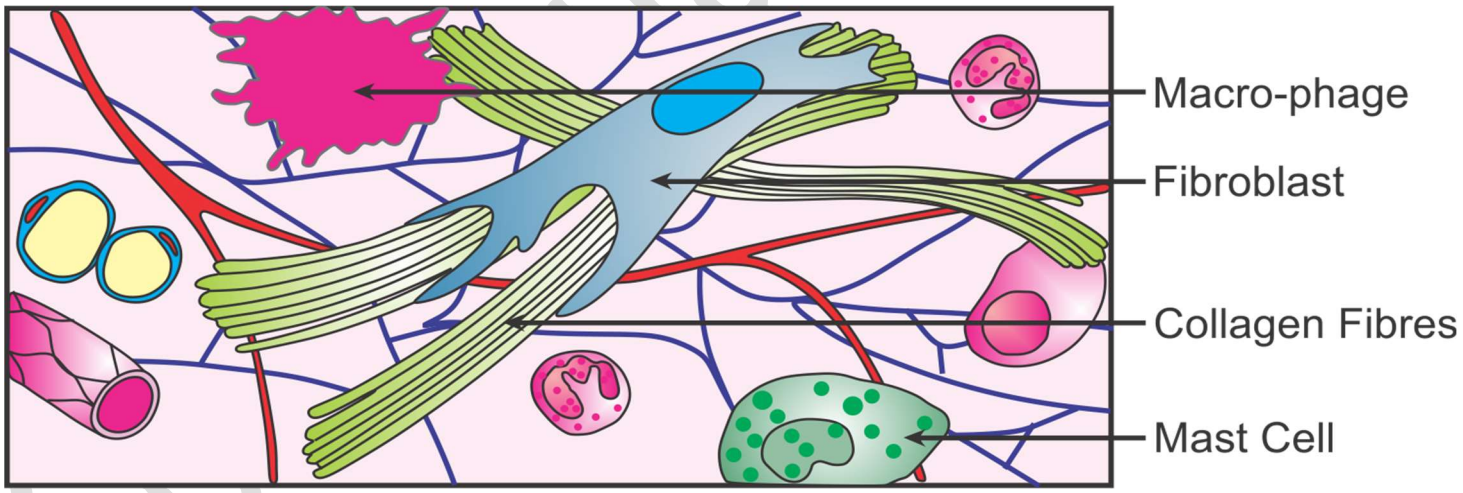
  ⇒ Loose Areolar Connective Tissue are tough yet flexible tissue that provides cushioning for many organs and tissues in the body.
- **Structure of Areolar Tissue:**

  ⇒ In this tissue fibers and cells are held together by a gel-like matrix. The cells of this tissue are called as Fibroblast.

  ⇒ Matrix has collagen and elastic fibers. Collagen fibers are a type of protein fibers that provides strength and cushioning to many areas.

  ⇒ Macrophages are white blood cells that protect the body's organs and structures by ingesting potential pathogens and cell debris in loose areolar connective tissue.

  ⇒ Mast cells are immune cells that are found in loose connective tissue. Mast cells play an important role in the immune system's response to certain bacteria and parasites, and they help control other types of immune responses.



⇒ **Location of Areolar Tissue:**

- Between Skin and Muscles.
- Around blood vessels and nerves.
- In bone marrow.
- In space inside organs.

⇒ **Functions of Loose Areolar Tissue:**

- It connects skin and muscles.
- It keeps the blood vessels in place.
- Helps in repair of tissues.

Conceptual Learning

- It supports internal organs.
- Fills the space inside organs.

- **Dense Areolar connective Tissue:**

⇒ Dense connective tissue proper is composed of a higher density of fibers, which may be regular or irregular or elastic.

⇒ Matrix of Dense connective tissue is dense or solids.

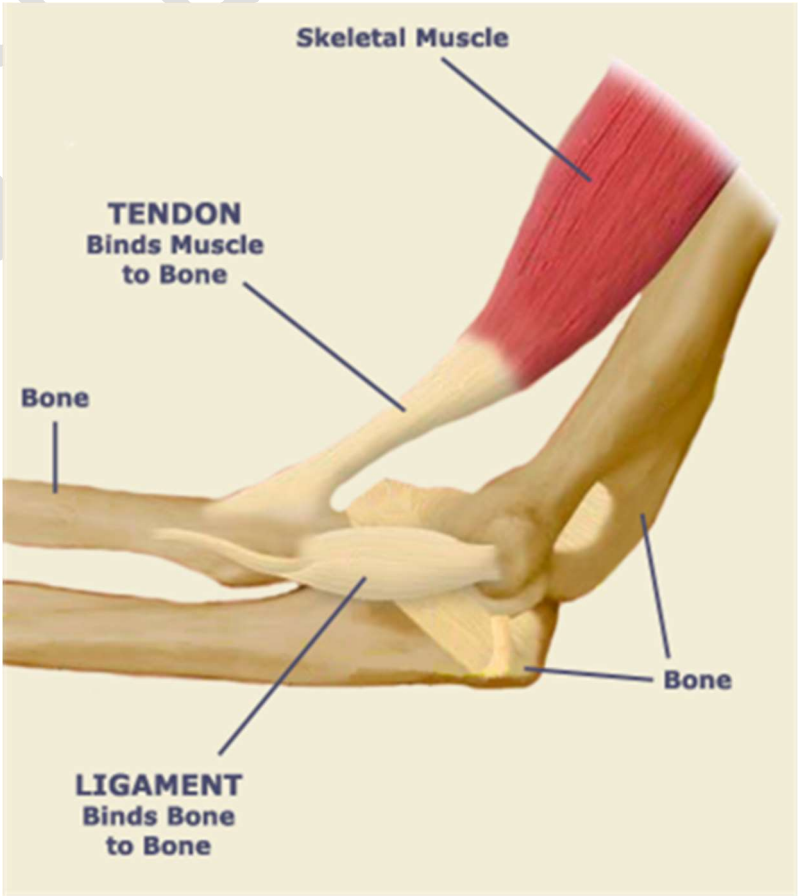
⇒ Cells of connective tissue are embedded in this dense matrix.

⇒ Types of Dense Areolar Connective Tissue:

i. Tendons ii. Ligaments

⇒ **Tendons**

- A tendon is a fibrous connective tissue that attaches muscle to bone.
- **Structure of Tendons**
  - It is a fibrous tissue.
  - It has a great strength.
  - It has cord-like structure.
  - It has a limited flexibility.
- **Functions of Tendons:**
  - They connect muscles to bones and transmits muscle forces to the bones and joints, helping muscles complete joint movements.
  - Tendons also help prevent muscle injury by absorbing some of the impact your muscles take when you run, jump, or do other movements.



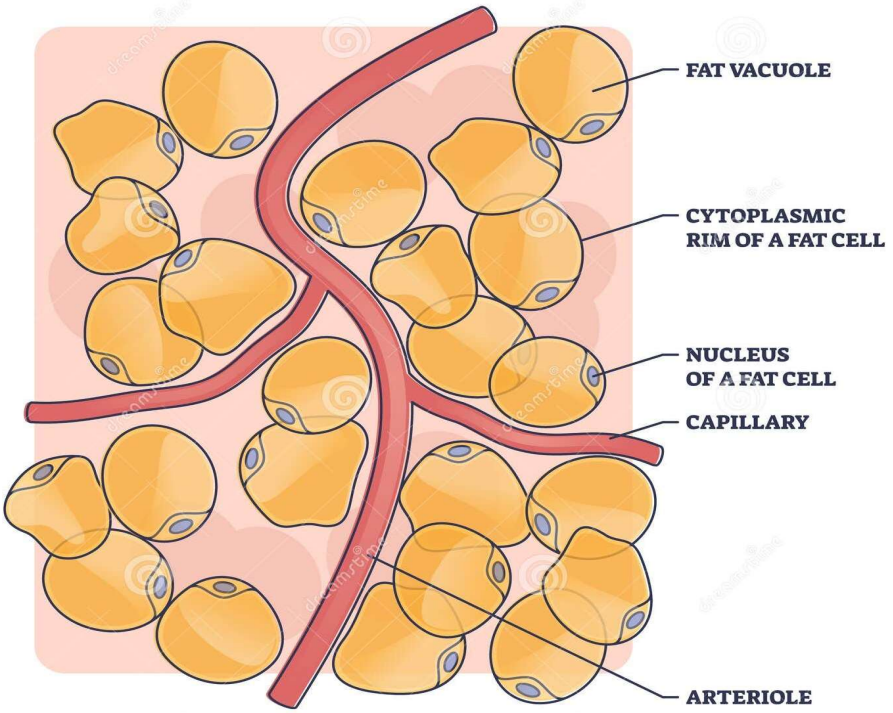
⇒ **Ligaments**

- Ligaments are bands of tough elastic tissue around your joints.
- **Structure of Ligaments:**
  - Ligaments are tough.
  - Ligaments are flexible bands.
  - It has a lot of strong collagen fibres.
  - Little matrix is present.
  - They are found in different shapes and sizes in the body.
- **Functions of Ligaments:**
  - They connect bones with bones in joins.
  - Some ligaments aren't connected to bones, but instead keep organs in place. For example, the broad ligament of the uterus keeps the womb in the right position in the pelvis.
  - Ligaments helps us to move properly.
  - Ligaments contain elastic fibers that allow them to stretch to some extent.

Conceptual Learning

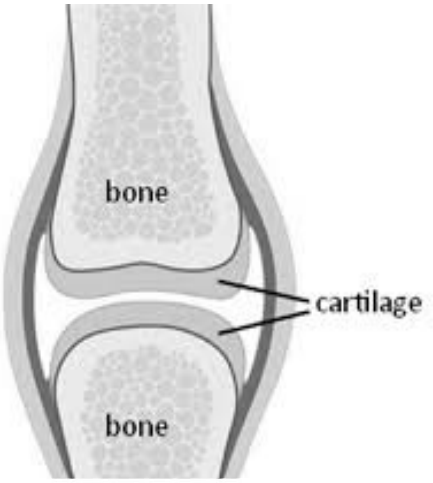
Adipose Connective Tissue:

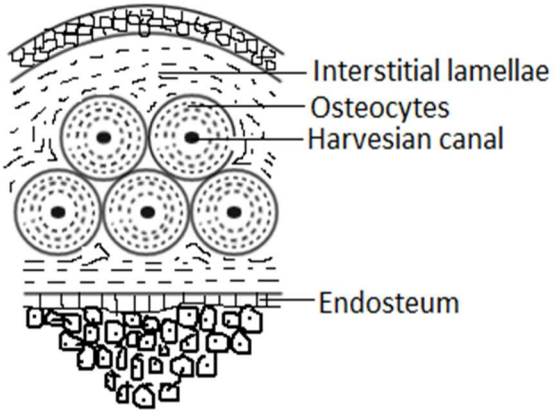
- Adipose tissue is a connective tissue made up of fat cells that stores energy.
- **Structure of Adipose connective tissue:**
  ⇒ Adipose tissue is an aggregation of fat cells.
  ⇒ Cells are round or oval.
  ⇒ Cells contains large fat droplets.
- **Functions of Adipose connective tissue:**
  ⇒ Storage of energy and release of energy.
  ⇒ It helps in temperature insulation (insulate the body from cold and heat).
  ⇒ Protects the kidneys.
- **Occurrence of Adipose connective tissue:**
  ⇒ It is found below skin.
  ⇒ It is found between internal organs.
  ⇒ Inner cavities of bone (Bone marrow adipose tissue).



Skeletal Connective Tissue:

- Skeletal connective tissue is a tissue that forms the endo-skeleton of vertebrates.
- Skeletal connective tissue is also known as supporting connective tissue.
- It is a type of connective tissue that's made up of cells and a solid matrix.
- **Types of Skeletal Connective Tissue:** There are two types of Skeletal Connective Tissues
  i. Bones ii. Cartilages
  ⇒ **Bones**
  - Bone forms the framework that supports the body.
  - It also anchors the muscles and supports the main organs of the body.
  - **Structure of Bone Cells:**
    - Bones are very strong, hard, rigid and non-flexible.
    - Bone cells are embedded in a hard matrix composed of calcium and phosphorus compounds and that why they are strong.
    - Bone cells are also called as **osteocytes.**
    - Haversian canals are microscopic tubes or tunnels in cortical bone that house nerve fibers and a few





Cross-section of compact Bone tissue

ConceptUa

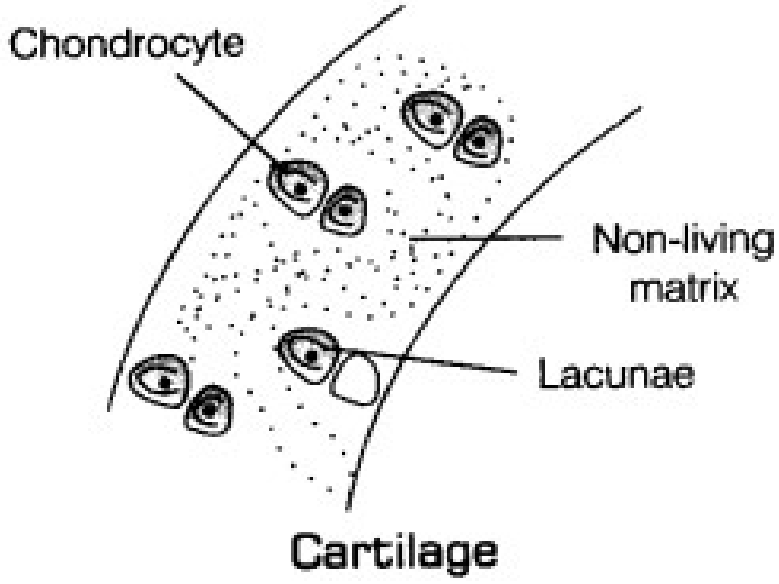
capillaries. This allows bone to get oxygen and nutrition without being highly vascular.

- Interstitial lamellae are irregular pieces of bone that fit into the gaps between the main bone units, called osteons. Interstitial lamellae provide rigid support to the bone
- Interstitial lamellae provide rigid support to the bone.

- **Functions of Bone:**
  - Provides shape and structure to the body.
  - Bones supports the body.
  - Protects vital body organs like brain, spinal cord and Lungs.
  - Bones helps in production of blood cells.
  - Helps in storage of minerals like calcium and phosphates.
- **Occurrence of Bone in Human body:**
  - Bones are present throughout the body.
  - Newborns have between 270 and 300 bones.
  - Adult human beings have 206 bones in their body.

⇒ **Cartilages**

- Cartilage is a strong, flexible connective tissue that protects your joints and bones.
- **Structure of Cartilages:**
  - It's made from specialized cells called **chondrocytes**.
  - Its cells are widely spaced out.
  - It has slightly elastic matrix made up of proteins and sugars which makes cartilages flexible.
  - It has collagen fibers.
  - Lacunae are small cavities in cartilage that house chondrocytes



- **Occurrence of Cartilages:**
  - Ear Pinna
  - Nose tip
  - Trachea
  - Larynx
  - Bone surfaces at Joints.
- **Functions of Cartilages:**
  - It provides support and flexibility to body parts.
  - It protects joints and bones by acting as shock absorber.
  - Provides structure and shape to some parts of body like respiratory tract.
  - Reduces friction and smoothens bone surface at joins.

Fluid Connective Tissue:

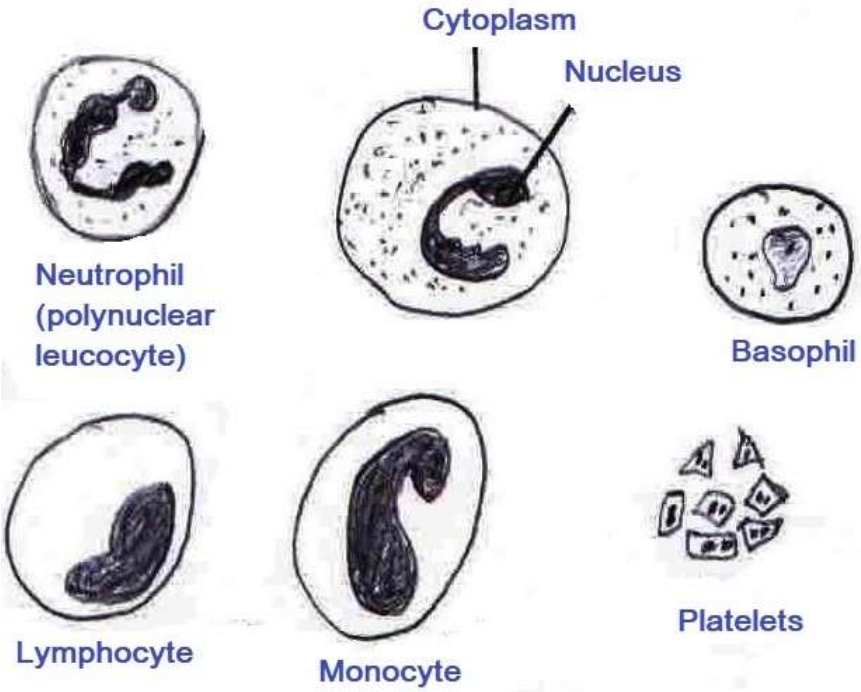
- Fluid connective tissue is also known as vascular connective tissue.
- It is the type of connective tissue that circulates various substances throughout the body**.**
- It Connects different parts of body.
- It maintains the continuity in the body.

Conceptual Learning

- **Types of Fluid Connective Tissue:** There are two types of Skeletal Connective Tissues
  ii. Blood
  iii. Lymph
- **Blood**

  ⇒ Blood is a specialized fluid that circulates throughout the body's blood vessels.

  ⇒ **Structure of Blood:**
  - It is composed of Blood Cells/Blood Corpuscles.
  - Matrix of blood cells is a liquid called plasma. The plasma contains proteins, salts and hormones.
  - There are three types of blood cells:
    - **RBC**: Red Blood Corpuscles
    - **WBC**: White blood Corpuscles
    - **Blood Platelets**
  - There are six types of WBC in blood as shown in figure.



  ⇒ **Functions of Blood:**
  - Transporting oxygen and nutrients to all the tissues.
  - Forming blood clots to prevent excess blood loss.
  - Carrying cells and antibodies that fight infection.
  - Bringing waste products to the kidneys which filter and clean the blood.
  - Regulating body temperature.
  - Transport of Hormones to various parts.

- **Lymph:**

  ⇒ Lymph, also known as lymphatic fluid.

  ⇒ Lymph is colorless, watery fluid that flows through the lymphatic system, which is part of the body's immune system.

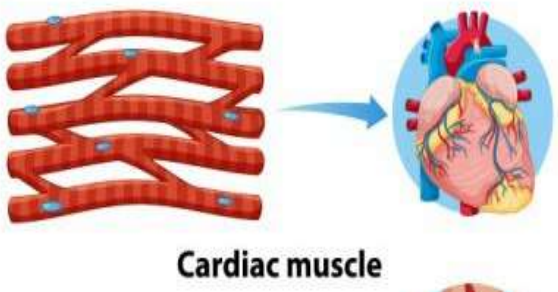
  ⇒ **Structure of Lymph:**
  - Its cells are called as Lymphocytes which are the type of white blood cells.
  - Its matrix is also plasma.

  ⇒ **Functions of Lymph:**
  - Deliver nutrients and proteins to cells of body.
  - Lymph protects your body from viruses, bacteria, and cancerous cells.
  - Lymph collects excess fluid from your tissues and returns it to your bloodstream, helping to maintain healthy fluid levels in your body.
  - Lymph absorbs fats from the small intestine through lymphatic vessels.
  - Lymph transports antibodies and lymphocytes to the blood.

Conceptual

✓ Muscular Tissue:

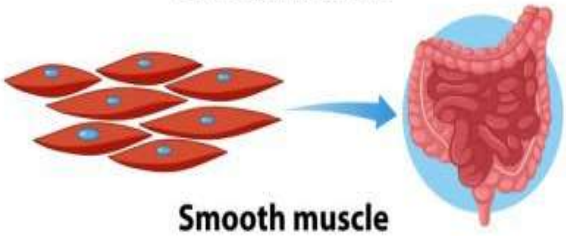
- Muscular tissue is a specialized tissue that helps muscles contract and move.
- Muscular tissue applies forces to different parts of the body by contraction.
- The Muscle tissue is highly cellular and is well supplied with blood vessels.
- **Characteristics of Muscular Tissue and its Cells:**
  - Muscle cells are also known as myocytes or muscle fibers.
  - Muscle cells are elongated and large in size.
  - Contractile proteins(Actin and Myosin) are present in fibers of muscle tissue which caused contraction and relaxation.



Cardiac muscle



Skeletal muscle



Smooth muscle

- **Functions of Muscular Tissue:**
  - Muscle tissue allows the body to move.
  - Muscle tissue supports the body's internal organs, such as those in the abdominal cavity.
  - Muscle tissue helps move food through the digestive system.
  - Muscle tissues help the heartbeat.
  - Muscle tissue helps empty the urinary bladder.
- **Types of Muscle Tissue:**
  i. Skeletal Muscle
  ii. Smooth Muscle
  iii. Cardiac Muscle
- **Skeletal Muscle:**
  - Skeletal muscles are the majority of muscles in the human body.
  - Skeletal muscles making up 30–40% of body weight and containing 50–75% of all body proteins.
  - They are attached to bones by tendons, which transmit the force of muscle contraction to the skeleton.
  - These muscles are attached to skeletal system and hence they are names as skeletal muscles.
  - It has striations (pattern of alternate light and dark bands) presents and thus called as striated muscles.
  - It generally allows voluntary actions (actions taken by consciousness) and thus called as voluntary Muscles.
  - **Structure of Skeletal Muscles**
    ⇒ Muscle fibers/Myocytes (Cells of skeletal Muscle) are elongated and cylindrical.
    ⇒ Skeletal muscle cells are longer than other types of muscle tissue.
    ⇒ Skeletal Muscle cells have multiple mitochondria to meet their energy needs.
    ⇒ Muscle cells are multinucleated in order to enhance oxygen uptake and tissue repair.
    ⇒ Skeletal muscles are unbranched because their muscle fibers are not divided into subdivisions.
    ⇒ Skeletal muscles appear striated or striped, because of the arrangement of actin and myosin proteins along the length of their muscle fibers.
  - **Functions of Skeletal Muscles**
    ⇒ It helps in body movement.
    ⇒ It helps in maintaining posture of body.
    ⇒ It helps to maintain the stability of joints.
    ⇒ Skeletal muscles protect your internal organs and tissues from injury.
    ⇒ Skeletal muscles store nutrients.
    ⇒ Skeletal muscles help regulate your body temperature.

Conceptual

Smooth Muscle:

- The smooth muscle is the type of muscle in the human body that is non-striated and performs involuntary actions (actions which are not performed by consciousness).
- Smooth muscles are the majority of muscles in the human body.
- These muscles cannot be started or stop by human will so they are called as Involuntary muscle.
- Smooth muscle tissue is generally found in the walls of hollow organs like intestines, uterus and stomach.
- These muscles do not get tired.
- **Structure of Smooth Muscles**
  - ⇒ Smooth muscle is made up of spindle-shaped(long with pointed ends) muscle fibers with a single nucleus.
  - ⇒ Smooth muscle fibers are shorter than skeletal muscle.
  - ⇒ Smooth Muscle fibers are uninucleate (single nucleus).
  - ⇒ They are unbranched.
  - ⇒ Smooth muscle fibers are non-striated as they their actin and myosin proteins are not arranged in regular manner along the cell fiber. Thus, they also called as Unstriated muscles.
  - ⇒ These are generally found in visceral organs(Soft internal organs with cavity of body), thus they are called as Visceral Muscle Tissue.
- **Functions of Smooth Muscles:**
  - ⇒ It propels food through the alimentary canal by undergoing coordinated contractions. The movement of food in the canal is called as peristaltic movement.
  - ⇒ Regulate the diameter of bronchioles and help the lungs expand when you breathe
  - ⇒ Dilate and constrict the pupil and change the shape of the lens.
  - ⇒ It regulates urine flow.
  - ⇒ It regulates blood pressure in blood vessels.

Cardiac Muscle:

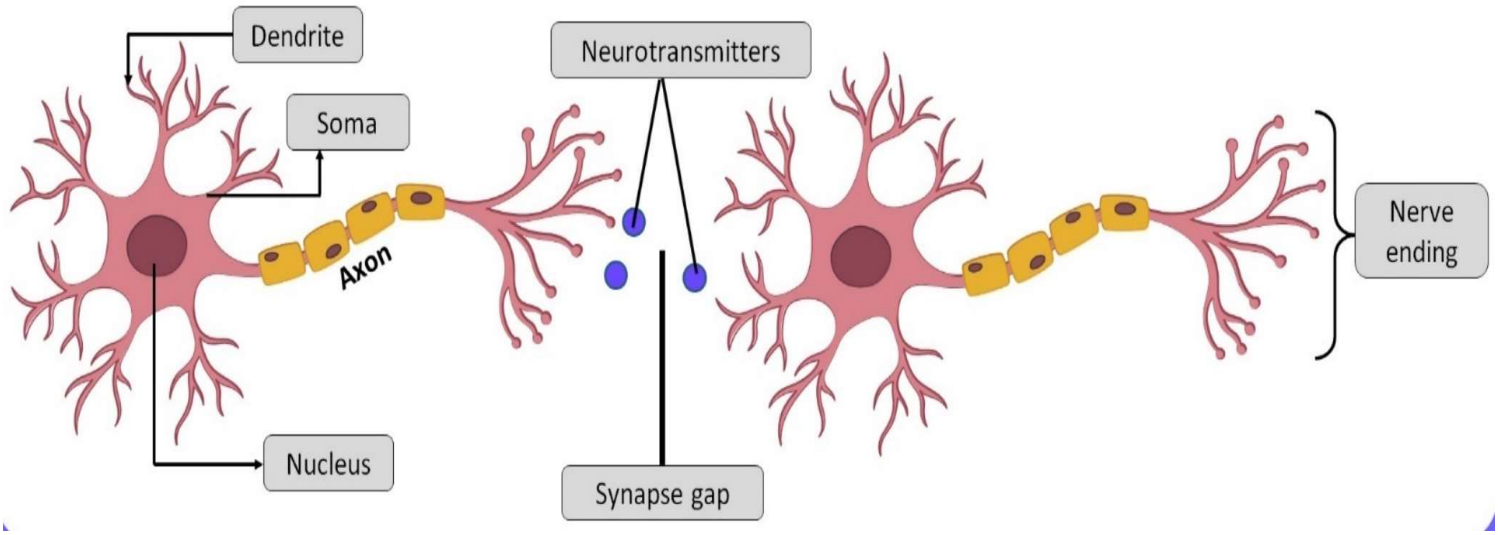
- Cardiac muscle is the middle layer of the heart that allows the heart to pump blood.
- These are only present in Heart.
- These muscles are also involuntary as they do not perform function according to our will.
- It has striations (pattern of alternate light and dark bands) presents.
- These muscles do not get fatigue(tiredness).
- **Structure of Cardiac Muscles:**
  - ⇒ Cardiac muscle cells are called as cardiomyocytes.
  - ⇒ Cardiac Muscle fibers are cylindrical in shape.
  - ⇒ Cardiac Muscle fibers are uninucleate.
  - ⇒ Cardiac muscle fibers are branched.
- **Function of Cardiac Muscles:**
  - ⇒ Cardiac muscle contracts and relaxes to pumps blood throughout the body's circulatory system.
  - ⇒ Cardiac muscle contractions create the force needed to push blood through arteries, which helps maintain blood pressure

✓ **Nervous Tissue:**

- Nervous tissue is also known as neural tissue.
- Nervous Tissue is the primary tissue in the nervous system, which regulates, controls and coordinate many body functions.

Conceptual Learning

- While all cells can respond to stimuli, nervous tissue cells are uniquely specialized to rapidly transmit these signals across the body.
- The brain, spinal cord and nerves are all composed of the nervous tissue.
- The cells of this tissue are called nerve cells or neurons.
- The nervous tissue is extremely delicate and can be damaged by the smallest amount of force.
- **Structure of Nervous Tissue:**
  - ⇒ The cells of Nervous tissue are called nerve cells or neurons.
  - ⇒ A neuron consists of a cell body with a nucleus and cytoplasm, from which long thin hair-like parts arise.
  - ⇒ **Parts of Neuron and their functions:**
  - **Cell Body (Soma):**
    - Nucleus: Contains the neuron's genetic material (DNA) and controls cellular activities, including growth and protein synthesis.
    - Cytoplasm: Houses various organelles
  - **Dendrites:**
    - Structure: Branch-like extensions that protrude from the cell body.
    - Function: Receive incoming signals from other neurons or sensory receptors. They have numerous receptors that bind neurotransmitters, converting chemical signals into electrical impulses.
  - **Axon:**
    - Structure: A long, thin projection extending from the cell body. Each neuron typically has one axon.
    - Function: Transmits electrical impulses (action potentials) away from the cell body to other neurons, muscles, or glands.
  - **Axon Terminals:**
    - Structure: The far ends of the axon branch out into tiny, bulb-like structures known as synaptic boutons or terminal.
    - Function: Release neurotransmitters (Chemicals especially Dopamine) into the synaptic cleft to communicate with other neurons, muscles, or gland cells.
  - ⇒ Gap between two neurons is called synapse.
  - ⇒ An individual nerve cell may be up to a meter long.
  - ⇒ Many neurons connect to form a nerve fiber and many nerve fibers connects to form a nerve/nervous tissue.
  - ⇒ The signal that passes along the nerve fiber is called a nerve impulse.



nceptual Learning

- **Functions of Nervous tissue: Nervous tissue has several key functions:**
  - Detecting Sensations: Senses changes in the environment, like touch, pain, and temperature.
  - Processing Information: Analyses the sensory input to understand what's happening and decide on a response.
  - Controlling Movements: Sends signals to muscles and glands to make them act, like moving your arm or sweating.
  - Maintaining Balance: Regulates internal conditions, such as body temperature and heart rate, to keep everything stable.
  - Learning and Memory: Helps you learn new things and remember information.
  - Emotional Responses: Influences feelings and behaviours, like happiness, fear, or anger.
  - Reflexes: Manages quick, automatic responses to certain stimuli, like pulling your hand away from something hot.

Conceptual Learning